

RIBSTONE SOUTH AGCM, AB, Canada

WMO: 710890

Lat: 52.5869N

Lon: 110.3428W

Elev: 2113

StdP: 13.61

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-24.8	-19.7	-31.0	1.0	-24.7	-25.8	1.4	-19.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.3	84.2	63.6	80.9	62.4	77.9	61.0	67.2	79.0	65.0	76.1	63.2	73.7	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	93.0	72.7	60.9	86.4	70.2	59.0	80.6	67.7	32.9	79.0	31.2	76.4	29.7	73.8	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-31.3	90.5	3.6	3.3	-33.8	92.8	-35.9	94.8	-38.0	96.6	-40.6	99.0		
(5)			-31.4	71.7	3.6	2.9	-34.0	73.8	-36.1	75.5	-38.1	77.2	-40.8	79.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.0	11.6	9.9	22.8	37.4	51.8	58.9	63.4	61.5	52.9	39.3	22.5	10.5		
	DBStd	22.62	15.95	14.48	13.82	9.53	7.64	5.57	4.61	5.66	8.13	9.17	12.89	14.40		
	HDD50	6081	1190	1124	843	387	71	6	0	1	62	348	824	1225		
	HDD65	10293	1655	1544	1308	829	412	193	87	137	368	796	1274	1690		
	CDD50	1345	0	0	0	8	125	274	415	357	149	17	0	0		
	CDD65	82	0	0	0	0	2	10	37	28	5	0	0	0		
	CDH74	1651	0	0	0	5	125	212	558	510	233	8	0	0		
	CDH80	366	0	0	0	0	16	30	125	120	75	0	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	15.2	0.7	0.4	0.6	1.1	1.6	2.8	2.8	2.0	1.1	0.7	0.6	0.6		
	PrecMax	20.6	1.5	0.9	1.4	3.2	3.4	4.7	5.4	5.0	2.9	1.6	2.0	1.1		
	PrecMin	9.5	0.2	0.2	0.0	0.1	0.2	0.9	1.0	0.2	0.0	0.1	0.1	0.2		
	PrecStd	3.0	0.4	0.2	0.3	0.7	0.9	1.1	1.0	1.0	0.9	0.4	0.5	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.4	42.7	55.3	71.3	82.1	83.7	87.8	87.9	88.2	73.3	55.0	42.3		
		MCWB	37.6	37.4	44.9	49.1	56.7	61.4	66.1	64.4	60.7	55.1	45.2	35.9		
	2%	DB	40.2	38.0	49.6	64.5	77.6	79.9	83.5	83.3	80.8	64.7	47.4	38.7		
		MCWB	35.5	34.2	40.7	47.1	55.0	60.6	66.1	63.8	59.1	50.6	40.6	34.3		
	5%	DB	37.6	35.0	44.1	59.2	73.1	76.3	80.0	79.7	74.9	59.1	42.2	34.5		
		MCWB	33.8	32.0	38.1	44.3	53.6	59.4	64.6	62.8	57.5	47.4	37.4	30.6		
	10%	DB	34.4	31.9	39.8	54.2	68.8	72.8	76.6	76.2	69.2	54.0	38.4	30.3		
		MCWB	31.3	29.4	35.1	42.2	51.5	57.7	62.8	61.8	54.8	44.9	34.8	27.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.1	37.9	46.0	51.0	58.7	65.8	72.0	69.2	63.1	56.5	45.7	36.9		
		MCDB	43.2	42.7	54.0	69.1	77.0	78.4	83.1	80.0	79.7	71.5	53.8	40.8		
	2%	WB	35.9	34.6	41.1	47.4	56.5	63.1	68.1	66.8	60.3	52.4	41.3	34.2		
		MCDB	39.9	37.5	48.0	62.9	72.9	74.7	80.0	79.0	76.9	61.3	46.5	38.4		
	5%	WB	33.9	32.3	38.3	45.1	54.8	61.2	66.0	64.5	57.9	48.6	37.7	30.6		
		MCDB	37.2	35.0	43.9	57.8	70.2	72.3	76.8	75.3	72.8	57.6	41.3	34.0		
	10%	WB	31.3	29.3	35.4	42.6	52.9	59.4	64.0	62.5	55.8	45.5	35.2	27.6		
		MCDB	34.3	31.9	39.5	52.8	67.0	70.1	74.1	73.2	68.1	53.4	38.4	30.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.9	19.0	18.5	21.4	25.5	22.3	23.3	24.2	25.1	21.0	16.0	16.4		
		MCDBR	17.5	19.2	21.9	31.0	32.1	28.9	30.1	32.2	36.1	29.9	21.1	19.5		
		MCWBR	13.9	16.0	14.7	15.8	14.4	12.9	14.8	15.0	16.8	17.0	14.5	15.7		
		MCDBR	17.9	19.4	21.4	29.0	29.3	24.3	26.9	26.9	33.0	28.1	19.3	19.0		
	5% WB	MCWBR	14.7	16.5	14.9	15.3	14.1	12.1	14.4	13.7	16.1	16.3	13.9	15.9		
(40) Clear-Sky Solar Irradiance	taub		0.248	0.289	0.289	0.336	0.359	0.373	0.367	0.379	0.319	0.283	0.258	0.235		
	taud		2.345	2.257	2.425	2.351	2.333	2.331	2.388	2.324	2.464	2.541	2.457	2.362		
	Ebn at Noon		245	261	290	285	283	279	279	267	275	267	242	231		
	Edh at Noon		18	27	28	35	37	38	35	36	27	20	16	15		
(44) All-Sky Solar Radiation	RadAvg		346	680	1144	1505	1837	1893	1998	1623	1148	683	345	258		
	RadStd		31	50	87	143	127	134	101	133	120	94	36	24		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page